

RPG FRAMEWORK

CUSTOM ENTITIES IN-GAME EDITOR

Complete UI design book

A Minecraft-native creator workflow for building, placing, testing, and debugging fully data-driven entities.

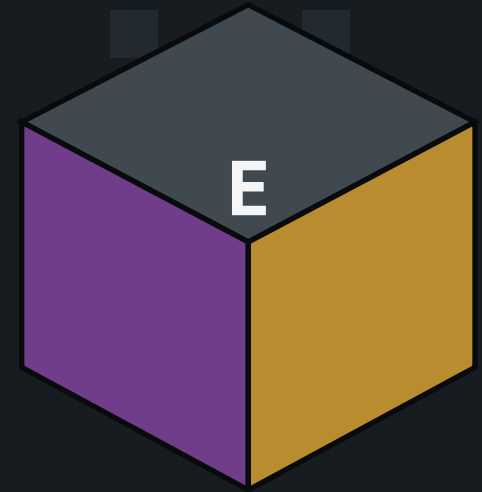
32 UI SCREENS

LIVE 3D PREVIEW

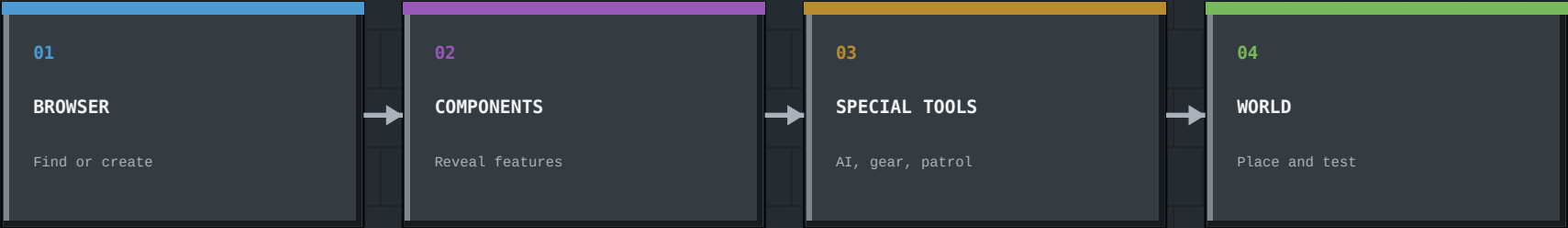
WORLD TOOLS

COMPANION TO CUSTOM STATS EDITOR

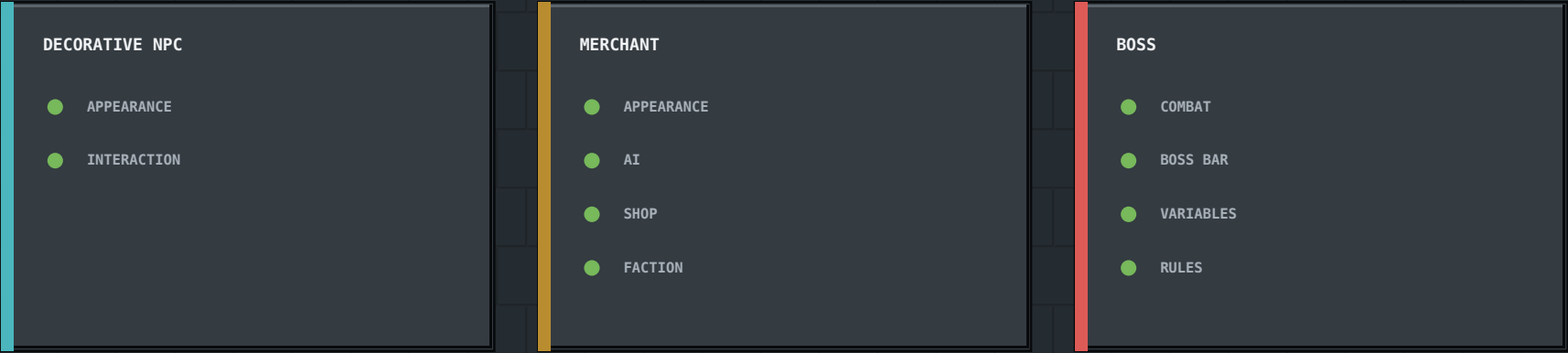
COMPONENTS / PREVIEW / SPECIALIZED TOOLS / NODE LOGIC



ENTITY WORKFLOW



VISIBLE COMPLEXITY



ONE BASE ENTITY CLASS / MANY DATA-DRIVEN DEFINITIONS / INSTANCE OVERRIDES STAY EXPLICIT

INTENT

Extend the shared RPG Framework workflow with components, a persistent entity preview, and in-world editing.

KEY STATES

Simple NPC, merchant, enemy, companion, boss, placed instance, and missing-addon definition.

BUILD NOTES

Reuse fields, pickers, graphs, validation, and dialogs from Custom Stats. Entity-specific tools own 3D preview and world selection.

CUSTOM ENTITIES

3 LOADED

+ NEW ENTITY



Search name or resource ID...

FILTER

All

IMPORT

RELOAD



VILLAGE BLACKSMITH

mypack:blacksmith

NPC

MERCHANT



BANDIT

mypack:bandit

HOSTILE



THE HOLLOW KING

mypack:hollow_king

BOSS

HOSTILE

BLACKSMITH

OPEN

SPAWN

DUPLICATE

RENAME

COPY ID

OPEN JSON

EXPORT

DELETE

3 RESULTS / FILTER: ALL

INTENT

Scan entity roles, components, and health at a glance, then open or act without extra screens.

KEY STATES

Selected, unsaved, invalid, recently edited, filtered, empty search, context menu, reload needed.

BUILD NOTES

Rows combine a head thumbnail with resource ID and role tags. Right-click actions place Delete last and Spawn near Open.

CREATE CUSTOM ENTITY

Choose a starting point. Everything remains editable.

NAME

Village Guard

RESOURCE ID

mypack:village_guard

Generated from name

TEMPLATE

Guard ▼

MODEL

Humanoid ▼

CANCEL

CREATE ENTITY

GUARD TEMPLATE

☒ APPEARANCE☒ MOVEMENT☒ COMBAT☒ ATTRIBUTES☒ AI☒ FACTION

INTENT

Create a working entity from four choices, while showing exactly what its template adds.

KEY STATES

Generated ID, invalid name, duplicate ID, template preview, model provider missing, create-ready.

BUILD NOTES

Name controls ID until manually edited. Templates add ordinary components and values, not different Java classes.

 VILLAGE BLACKSMITH

DEFINITION

UNSAVED

SPAWN

SAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

GENERAL

Definition identity and player-facing metadata.

DISPLAY NAME

Village Blacksmith

RESOURCE ID

mypack:blacksmith

DESCRIPTION

The village's local blacksmith and merchant.

COMPONENTS

APPEARANCE

Required

...

SHOP

Enabled

...

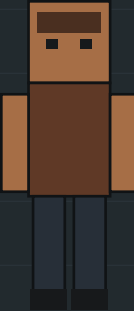
FACTION

Enabled

...

LIVE PREVIEW

HUMANOID



DRAG TO ROTATE

IDLE

WALK

ATTACK

100%

50%

150%

LMB ROTATE / WHEEL ZOOM

INTENT

Keep entity capabilities on the left, the selected form in the center, and a live 3D preview always visible.

KEY STATES

Component enabled, selected, missing dependency, removable, required, Basic/Advanced visibility.

BUILD NOTES

Components own navigation pages and serialization. Required components cannot be removed; unknown ones keep raw data.

ADD COMPONENT

Search components...

ALL 16

BEHAVIOR 5

RPG 5

WORLD 3

ADVANCED 3

☒ Compatible only

AI

Goals, targeting, schedules

ADDED

COMBAT

Melee, ranged, custom attacks

ADD

DIALOGUE

Open RPG Framework dialogue

ADD

SHOP

Merchant inventory and pricing

ADD

FACTION

Allies, enemies, reputation

ADD

BOSS

Boss bar and presentation

ADD

CANCEL

ADD SELECTED

INTENT

Expose registered entity capabilities through a searchable catalog that scales to addons.

KEY STATES

Already added, incompatible, missing dependency, addon badge, keyboard selection, generic fallback editor.

BUILD NOTES

Registry metadata supplies category, icon, dependencies, conflicts, defaults, and optional custom editor widget.

VILLAGE BLACKSMITH

DEFINITION

UNSAVED

SPAWN

SAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

GENERAL

DISPLAY NAME

Village Blacksmith

RESOURCE ID

mypack:blacksmith

Referenced by 3 quests, 2 dialogue, 1 spawn rule.

VIEW REFERENCES

TAGS

VILLAGE

MERCHANT

+

☒ Visible nameplate

DISTANCE

32 blocks

DESCRIPTION

The village's local blacksmith...

LIVE PREVIEW

HUMANOID

IDLE

WALK

ATTACK

100%

50%150%

LMB ROTATE / WHEEL ZOOM

INTENT

Keep definition identity, tags, and nameplate behavior together with reference-aware ID changes.

KEY STATES

Tag picker, hidden nameplate, invalid ID, duplicate ID, referenced rename, migration preview.

BUILD NOTES

Tags use free-form resource-safe tokens. Reference counts appear before the creator commits an ID change.

VILLAGE BLACKSMITH

DEFINITION

UNSAVED

SPAWN

SAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

APPEARANCE

MODEL

Humanoid

SCALE

1.00

TEXTURE

mypack:blacksmith.png

BROWSE

HITBOX

WIDTH

0.60

HEIGHT

1.95

EYE HEIGHT

Auto

SHADOW

0.50

TINT

#FFFFFF

LAYER

Cutout

☒ Render armor

☒ Render held items

☐ Glow outline

☒ Show hitbox preview

PREVIEW STATE

Idle / Day / Variant 1

LIVE PREVIEW

HUMANOID

IDLE

WALK

ATTACK

100%

50%150%

LMB ROTATE / WHEEL ZOOM

INTENT

Make every visual change immediate and keep model, texture, proportions, and render options together.

KEY STATES

Missing texture, incompatible model, transparent/glowing render, custom hitbox, animation preview.

BUILD NOTES

Preview updates locally before save. Hitbox and eye-height overlays can be toggled directly in the viewport.

APPEARANCE / VISUAL ASSETS

DONE

MODEL PROVIDER



Search models...

BUILT-IN

HUMANOID

ZOMBIE

SKELETON

VILLAGER

QUADRUPED

CUSTOM

MYPACK:GOLEM

MYPACK:HOLLOW_KING

ADDONS

CUSTOM ANIMATION MODEL

TEXTURE VARIANTS

+ ADD VARIANT



blacksmith_1.png

Weight

5

PREVIEW

X



blacksmith_2.png

Weight

2

PREVIEW

X



blacksmith_old.png

Weight

1

PREVIEW

X

VARIANT PREVIEW

PREV

NEXT

RANDOMIZE

1 / 3

INTENT

Manage model providers and weighted texture variation without leaving the appearance workflow.

KEY STATES

Built-in/custom/addon provider, missing provider, zero weight, duplicate texture, random preview.

BUILD NOTES

Friendly names remain paired with resource IDs. Randomize uses normalized weights and explains invalid totals.

 VILLAGE BLACKSMITH

DEFINITIONUNSAVED

SPAWNSAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

ATTRIBUTES

+ ADD ATTRIBUTE

MAX HEALTH

30

X

MOVEMENT SPEED

0.22

X

ATTACK DAMAGE

4

X

ARMOR

2

X

FOLLOW RANGE

24

X

DAMAGE MODIFIERS

FIRE DAMAGE

X 2.0

PROJECTILE

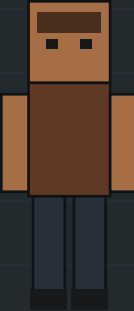
X 1.5

FALL DAMAGE

IMMUNE

LIVE PREVIEW

HUMANOID



DRAG TO ROTATE

IDLE

WALK

ATTACK

100%

50%150%

LMB ROTATE / WHEEL ZOOM

INTENT

Show only explicitly configured attributes, then expose damage modifiers as a focused advanced section.

KEY STATES

Inherited default, invalid range, duplicate attribute, immunity, conditional modifier, unknown attribute.

BUILD NOTES

Attribute pickers come from the runtime registry. Empty definitions inherit base defaults and stay visually distinct.



VILLAGE BLACKSMITH

DEFINITION

UNSAVED

SPAWN

SAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

MOVEMENT

MOVEMENT TYPE

Ground

SPEED

Use movement attribute

CAPABILITIES



Can jump



Can swim



Open doors



Avoid water



Can fly



Climb ladders

STEP HEIGHT

Default

TURN SPEED

Vanilla

GROUND SETTINGS VALID

LIVE PREVIEW

HUMANOID



DRAG TO ROTATE

IDLE

WALK

ATTACK

100%

50%

150%

LMB ROTATE / WHEEL ZOOM

INTENT

Reveal settings that match the selected movement mode instead of showing every locomotion option.

KEY STATES

Ground, flying, swimming, hovering, stationary, inherited speed, impossible option combination.

BUILD NOTES

Changing mode preserves inactive settings in memory but serializes only fields supported by the active mode unless requested.



VILLAGE BLACKSMITH

DEFINITION

UNSAVED

SPAWN

SAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

AI GOALS

+ ADD GOAL

::

1



MELEE ATTACK

Speed 1.0 / Keep target

EDIT

::

2



RETURN TO HOME

Outside 24 blocks

EDIT

::

3



RANDOM WANDER

Speed 0.8 / Radius 12

EDIT

::

4



LOOK AT PLAYER

Within 8 blocks

EDIT

::

5



RANDOM LOOK

Default interval

EDIT

TARGETING IS CONFIGURED SEPARATELY

LIVE PREVIEW

HUMANOID



DRAG TO ROTATE

IDLE

WALK

ATTACK

100%

50%

150%

LMB ROTATE / WHEEL ZOOM

INTENT

Represent normal behavior as a readable ordered list, with drag priority and plain-language summaries.

KEY STATES

Drag reorder, duplicate goal, disabled, invalid, no goals, active-at-runtime indicator, edit conditions.

BUILD NOTES

Lower priority number runs first. A drag commits one undo action; runtime-active highlighting is debug-only.

RANDOM WANDER

SPEED

0.8

INTERVAL

120 ticks

MAXIMUM DISTANCE

12 blocks

CONDITIONS

No conditions

+ ADD

CANCEL

SAVE GOAL

TARGET SELECTION

+ ADD RULE

1 RETALIATE WHEN ATTACKED

Any attacker

EDIT

2 ATTACK FACTION ENEMIES

Faction: Village

EDIT

3 ATTACK PLAYER

Bandit Reputation <= -25

EDIT

Target rules choose candidates; AI goals act on them.

INTENT

Keep 'what should I do?' separate from 'who should I target?' while sharing the same condition builder.

KEY STATES

Goal with conditions, invalid interval, retaliation, faction enemy, conditional player target, no selector.

BUILD NOTES

Goal editors are schema-driven. Target rules are evaluated independently and can be reused by melee or ranged goals.

 VILLAGE BLACKSMITH

DEFINITION

UNSAVED

SPAWN

SAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

AI / HOME & PATROL

HOME

X

Y

Z

RADIUS

124

68

-42

16

USE CURRENT POSITION

SELECT IN WORLD

PATROL

MODE

Loop

EDIT IN WORLD

1

WEST GATE

Wait 40

>

2

MARKETPLACE

Wait 100

>

3

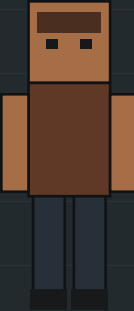
EAST GATE

Wait 40

>

LIVE PREVIEW

HUMANOID



DRAG TO ROTATE

IDLE

WALK

ATTACK

100%

50%

150%

LMB ROTATE / WHEEL ZOOM

INTENT

Configure spatial behavior precisely while offering world selection for coordinates and waypoints.

KEY STATES

Current position, invalid dimension, no home, loop/reverse/random/once, waypoint action, empty patrol.

BUILD NOTES

Positions include dimension IDs. World selection returns to this screen with a preview before committing.

PATROL EDIT / VILLAGE GUARD

POINT 2 SELECTED

CANCEL

SAVE

CONTROLS

LMB ADD / RMB REMOVE / WHEEL SELECT

ENTER SAVE / ESC CANCEL

MARKETPLACE

Wait 100 ticks

Facing: East

INTENT

Place and adjust waypoints directly in the world with an always-visible, reversible control guide.

KEY STATES

Add, remove, select, reorder, invalid surface, cross-dimension point, cancel, save, unsaved world edit.

BUILD NOTES

Render numbered markers, connecting lines, arrival facing, and wait time. Escape restores the pre-edit patrol.

VILLAGE BLACKSMITH

DEFINITION

UNSAVED

SPAWN

SAVE

- GENERAL
- APPEARANCE
- ATTRIBUTES
- MOVEMENT
- AI
- COMBAT
- EQUIPMENT
- INTERACTION

MORE COMPONENTS

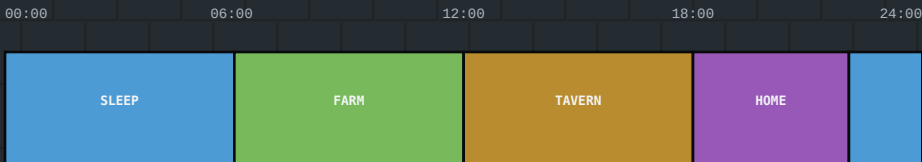
DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

NPC SCHEDULE

+ ADD ENTRY



- 06:00

MOVE TO FARM

Farm plot

EDIT
- 12:00

MOVE TO TAVERN

Village tavern

EDIT
- 18:00

MOVE HOME

Blacksmith home

EDIT
- 22:00

SLEEP

Until 06:00

EDIT

LIVE PREVIEW

HUMANOID



IDLE

WALK

ATTACK

100%

50%

150%

LMB ROTATE / WHEEL ZOOM

INTENT

Turn daily routines into a readable 24-hour timeline with precise entry configuration.

KEY STATES

Overlapping entry, empty period, cross-midnight sleep, conditional schedule, shop open/closed preview.

BUILD NOTES

Timeline drag changes start time; exact numeric entry is always available. Actions use the shared action registry.



VILLAGE BLACKSMITH

DEFINITION

UNSAVED

SPAWN

SAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

EQUIPMENT

DRAG ITEMS OR CLICK A SLOT

HEAD



IRON HELMET

MAIN HAND



OFF HAND



CHEST



LEATHER APRON

IRON AXE

LEGS



IRON LEGS

SELECTED SLOT

☒ Render item

FEET



LEATHER BOOTS

DROP CHANCE

8.5%

WEIGHTED ALTERNATIVES

CLEAR SLOT

LIVE PREVIEW

HUMANOID



IDLE

WALK

ATTACK

100%

50%

LMB ROTATE / WHEEL ZOOM

150%

INTENT

Use a familiar Minecraft inventory metaphor for visible gear, random sets, and drop chances.

KEY STATES

Drag item, empty slot, weighted alternatives, hidden rendering, invalid item, drop chance, clear slot.

BUILD NOTES

Accept real item stacks where permissions allow; otherwise use the shared item picker and preserve components.

COMBAT CONFIGURATION

RANGED ACTIVE

DONE

COMBAT TYPE

☐ None ☐ Melee ☒ Ranged ☐ Custom

MELEE SETTINGS

SAVED

ATTACK RANGE

2.5 blocks

COOLDOWN

20 ticks

MOVE SPEED

1.0

KNOCKBACK

0.0

☒ Keep target after hit

☒ Require line of sight

Settings preserved when another type is active.

RANGED SETTINGS

ACTIVE

PROJECTILE

Arrow (minecraft:arrow)

RANGE

20

COOLDOWN

30 ticks

SPEED

1.5

ACCURACY

90%

BURST

1

SPREAD

0.0

☒ Lead moving targets

☒ Line of sight

WARNING: no player target rule

INTENT

Switch between none, melee, ranged, and custom combat while exposing only relevant fields.

KEY STATES

Type change, missing projectile, invalid burst, no target selector, custom attack reference, cooldown conflict.

BUILD NOTES

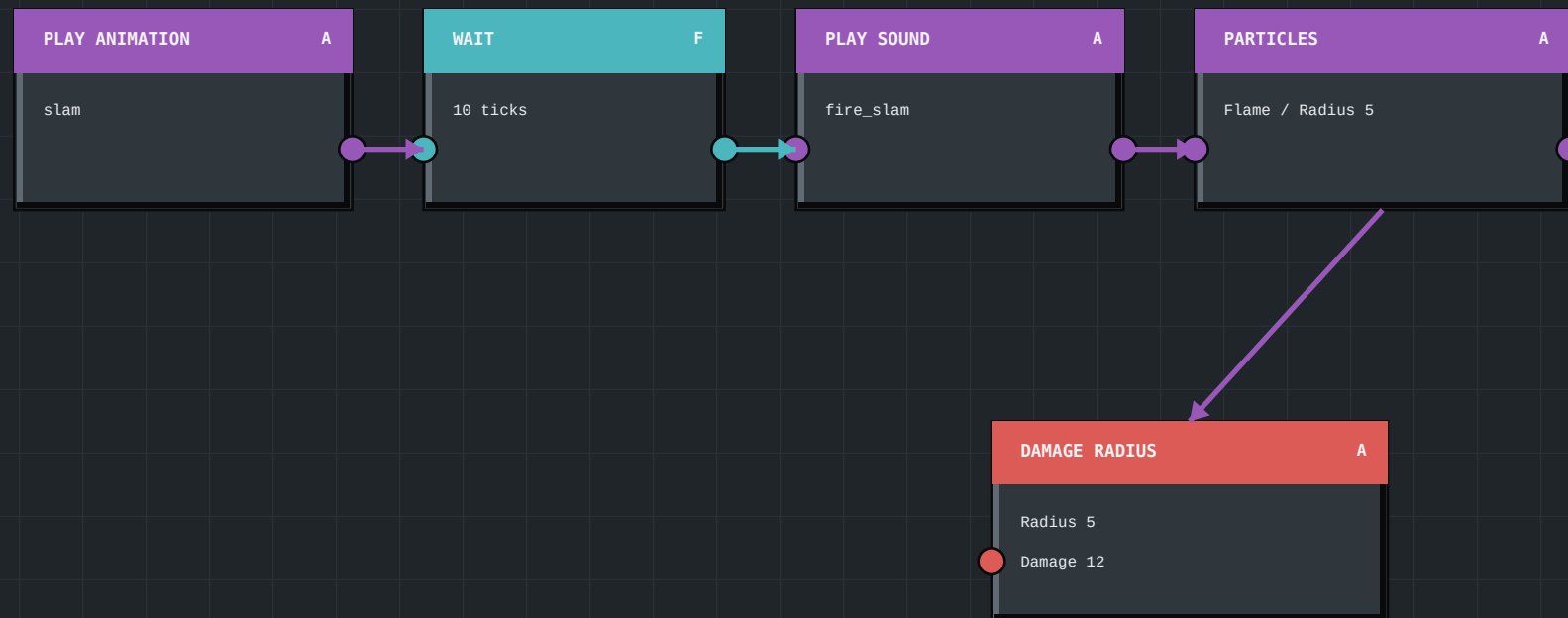
Combat describes attacks; targeting remains in AI. Ranged projectile pickers support friendly labels and raw IDs.

CUSTOM ATTACK / FIRE SLAM

UNSAVED

TEST

DONE



TAB ADD NODE

SPACE PAN / WHEEL ZOOM

INTENT

Compose cinematic or multi-step attacks with the shared action graph without forcing graphs on basic combat.

KEY STATES

Wait, animation, sound, particles, radial damage, branch, interrupted attack, missing animation, cooldown.

BUILD NOTES

Sequence ports are typed flow. Attack definitions become reusable resources referenced by AI attack goals.

 VILLAGE BLACKSMITH

DEFINITIONUNSAVED

SPAWNSAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

INTERACTIONS

+ ADD

RIGHT CLICK

OPEN DIALOGUE

mypack:blacksmith_intro

EDIT >

SNEAK + RIGHT CLICK

OPEN SHOP

mypack:blacksmith_shop

EDIT >

RIGHT CLICK

DIALOGUE IF QUEST COMPLETE

mypack:blacksmith_thanks

IF LOST_HAMMER COMPLETE

EDIT >

LINKED OBJECTS

DIALOGUE

blacksmith_intro

SHOP

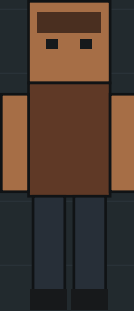
blacksmith_shop

FACTION

village

LIVE PREVIEW

HUMANOID



DRAG TO ROTATE

IDLE

WALK

ATTACK

100%

50%150%

LMB ROTATE / WHEEL ZOOM

INTENT

Keep common trigger-condition-action interactions readable and make linked RPG resources one click away.

KEY STATES

Right-click, sneak modifier, condition, multiple actions, missing dialogue/shop, edit linked object, complex graph.

BUILD NOTES

Resource picker results include type, namespace, validation state, and Edit action. Graph mode is opt-in per interaction.



VILLAGE BLACKSMITH

DEFINITION

UNSAVED

SPAWN

SAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

DROPS & SOUNDS

DROPS

LOOT TABLE

mypack:entities/bandit

MINECRAFT XP

10

RPG XP

25

EDIT LOOT TABLE

SOUNDS

AMBIENT

entity.villager.ambient



HURT

entity.villager.hurt



DEATH

entity.villager.death



ATTACK

None



INTERACT

None



> PREVIEW SOUND

LIVE PREVIEW

HUMANOID



DRAG TO ROTATE

IDLE

WALK

ATTACK

100%

50%

150%

LMB ROTATE / WHEEL ZOOM

INTENT

Reuse Minecraft-native loot and sound resources while adding RPG XP and immediate audio preview.

KEY STATES

Missing loot table, no drops, invalid sound, preview unavailable on server, custom XP, weighted loot editor jump.

BUILD NOTES

Do not duplicate loot-table functionality. Sound preview plays client-side and clearly labels source IDs.

SPAWNING / BANDIT

DONE

☒ Can spawn naturally

WEIGHT

10

GROUP MIN

2

GROUP MAX

5

DIMENSIONS

minecraft:overworld

BIOMES

#minecraft:is_forest

LIGHT MIN

0

LIGHT MAX

7

HEIGHT MIN

-64

HEIGHT MAX

320

CONDITIONS

TIME: NIGHT

DIFFICULTY >= NORMAL

+ ADD CONDITION

SPAWN TEST ENTITY

SPAWN TEST GROUP

SPAWN TEST

Current position: 124, 68, -42



Correct biome

minecraft:forest



Correct dimension

overworld



Light level too high

12 > 7



Height valid

68



Time valid

Night

ENTITY CANNOT SPAWN HERE

INTENT

Configure natural spawning in compact sections and explain exactly why a test location fails.

KEY STATES

Natural off, missing dimension, group min > max, biome tag, custom condition, successful/failed test.

BUILD NOTES

Test evaluates registered spawn conditions at the creator's current position and never spawns unless requested.

 VILLAGE BLACKSMITH

DEFINITIONUNSAVED

SPAWNSAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

PERSISTENCE

DESPAWN

Never

Save instance state

Unique entity

UNIQUE ENTITY

UNIQUE ID

village_blacksmith

Prevent duplicate spawn

RESPAWNING

Next day at home

RESPAWN DELAY

1 day

SPAWN ANCHOR

Home

UNIQUE ID AVAILABLE

LIVE PREVIEW

HUMANOID



DRAG TO ROTATE

IDLE

WALK

ATTACK

100%

50%150%

LMB ROTATE / WHEEL ZOOM

INTENT

Make despawn, instance saving, uniqueness, and respawning explicit for story NPCs and bosses.

KEY STATES

Vanilla-like despawn, never, distance, delayed respawn, duplicate unique ID, missing spawn anchor.

BUILD NOTES

Unique enforcement is server-authoritative. Warn before enabling uniqueness on definitions already placed multiple times.

HOLLOW KING / ADVANCED BEHAVIOR

DONE

VARIABLES

+ ADD

phase

INTEGER

1

Used by 2 rules

angry

BOOLEAN

false

Used by 2 rules

player_met

BOOLEAN

false

Used by 2 rules

ritual_progress

INTEGER

0

Used by 2 rules

RULES

+ NEW RULE

PHASE TWO

OPEN >

IF

Health <= 50%

DO

Set phase = 2 / Summon minions

BECOME ANGRY

OPEN >

IF

Player attacks entity

DO

angry = true

NIGHT DIALOGUE

OPEN >

IF

Time >= 13000

DO

Set dialogue night_dialogue

INTENT

Expose per-instance state and summarize behavior in plain language before opening the graph.

KEY STATES

Duplicate variable, unknown type, variable reference count, invalid rule, disabled rule, empty state.

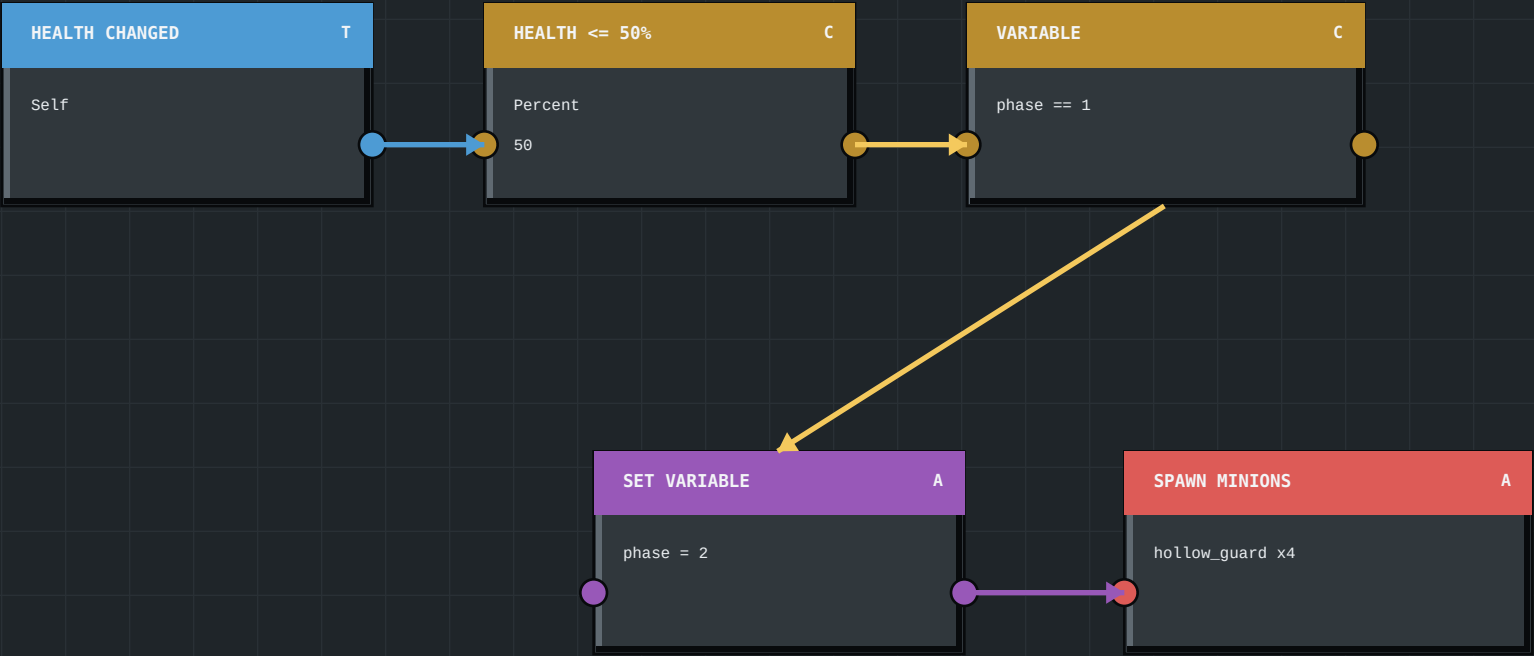
BUILD NOTES

Variables become available automatically in conditions, actions, dialogue, interactions, and boss logic.

RULE / PHASE TWO

UNSAVED

DONE



TRACE PREVIEW

Trigger > Health > phase > 2 actions

INTENT

Reuse the shared node language for variables, health, actions, and complex branches.

KEY STATES

Missing variable, disconnected trigger, unknown addon node, typed connection, debug trace, unsaved graph.

BUILD NOTES

Entity nodes extend the same registry used by stats and quests. Unknown nodes remain visible with preserved payload.

 THE HOLLOW KING

DEFINITION

UNSAVED

SPAWN

SAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

BOSS

☒ Boss bar enabled

BOSS NAME

The Hollow King

COLOR

Purple

STYLE

Progress

VISIBILITY

64 blocks

BOSS MUSIC

hollow_king_theme

☒ Darken sky

☐ Create fog

BAR PREVIEW

THE HOLLOW KING

216 / 300

PHASES ARE BUILT IN VARIABLES + RULES

LIVE PREVIEW

HUMANOID



DRAG TO ROTATE

IDLE

WALK

ATTACK

100%

50%150%

LMB ROTATE / WHEEL ZOOM

INTENT

Add boss presentation without creating a separate entity type or duplicating phase logic.

KEY STATES

Bar off, custom name, color/style, visibility, dark sky, fog, missing music, low-health warning.

BUILD NOTES

Boss phases remain variables plus rules plus AI. This component owns presentation and discovery range only.

 VILLAGE BLACKSMITH

DEFINITION

UNSAVED

SPAWN

SAVE

GENERAL

APPEARANCE

ATTRIBUTES

MOVEMENT

AI

COMBAT

EQUIPMENT

INTERACTION

MORE COMPONENTS

DROPS SPAWNING

RULES ADVANCED

+ ADD COMPONENT

ADVANCED

FILE

data/mypack/rpg/entities/blacksmith.json

VIEW JSON

RELOAD DISK

RUNTIME

JSON PREVIEW

```
{
  "id": "mypack:blacksmith",
  "appearance": {
    "model": "rpg:humanoid",
    "texture": "mypack:blacksmith.png"
  },
  "attributes": { "minecraft:max_health": 30 }
}
```

COPY

EDIT RAW

VALID

REFERENCES17 objectsVIEW >

LIVE PREVIEW

HUMANOID



DRAG TO ROTATE

IDLE

WALK

ATTACK

50%100%150%

LMB ROTATE / WHEEL ZOOM

INTENT

Expose source file, references, validation, and runtime inspection without crowding normal creation.

KEY STATES

Read-only JSON, raw edit, reload conflict, missing file, unknown fields preserved, runtime definition mismatch.

BUILD NOTES

Read-only is the default. Raw edits require parse and schema validation with a recoverable last-valid buffer.

EDITING INSTANCE / WEST GATE GUARD

INSTANCE

RESET

SAVE

DEFINITION

mypack:village_guard

EDIT DEFINITION

INSTANCE UUID 3c18...a8f2

OVERRIDES

+ ADD OVERRIDE

NAME

West Gate Guard

RESET

HOME

West Gate / 124, 68, -42

RESET

PATROL

west_gate_patrol

RESET

DIALOGUE

mypack:gate_guard

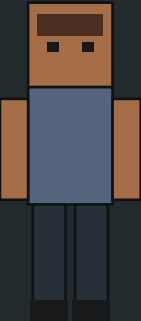
INHERITED

FACTION

mypack:village

INHERITED

INSTANCE PREVIEW



WEST GATE GUARD

DIAMOND DEFINITION

CIRCLE OVERRIDE

INTENT

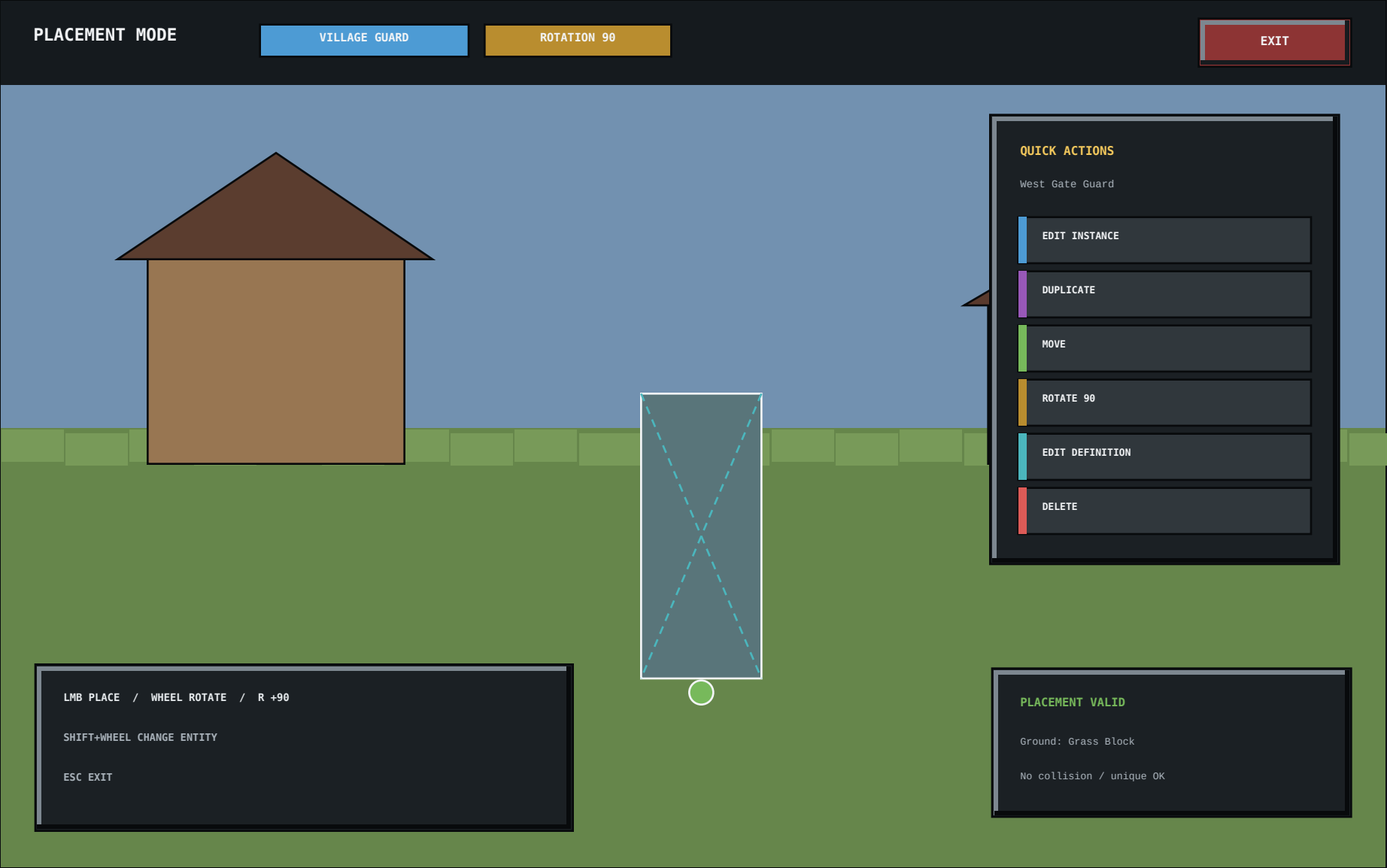
Make it impossible to confuse editing one placed NPC with changing every entity using the definition.

KEY STATES

Definition value, instance override, reset override, unsupported override, UUID, stale definition, edit definition.

BUILD NOTES

Blue diamonds mean definition values; gold circles mean per-instance overrides. Save writes only changed overrides.



INTENT

Place reusable definitions quickly with a transparent preview that communicates rotation, scale, hitbox, and grounding.

KEY STATES

Valid/invalid ground, rotate, cycle entity, duplicate, quick actions, move existing, protected delete, exit.

BUILD NOTES

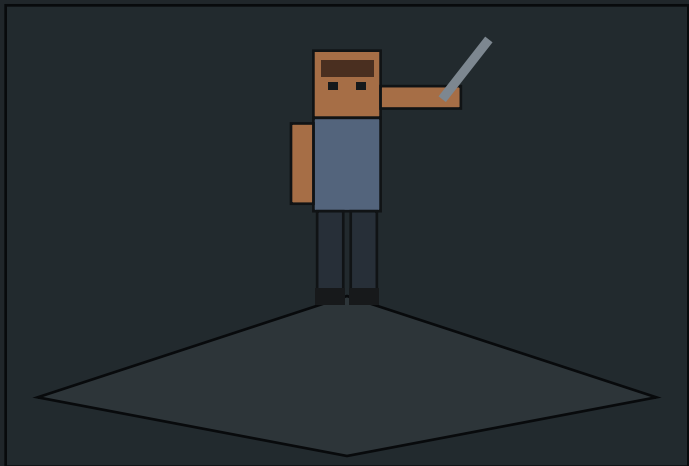
Preview placement is client-side; the server validates permission, collision, uniqueness, and final position on click.

ENTITY DEBUG / VILLAGE GUARD

TEST ENTITY

RESET TEST

DELETE



HEALTH



TARGET Zombie #458

CURRENT GOAL **MELEE ATTACK**

ANIMATION attack_swing

RUNTIME STATE

NAVIGATION

Status	Moving
Target	130, 65, -21
Path nodes	8

FACTION

Current	Village
Relation	Zombie: Enemy

VARIABLES

angry	false
phase	1
player_met	true

RULES & COOLDOWNS

Active rules	3
Melee cooldown	7 ticks

INTENT

Shorten iteration by spawning a temporary entity and exposing current AI, navigation, variables, attacks, and animations.

KEY STATES

No test entity, dead/unloaded, paused AI, selected target, path failure, active cooldown, reset/delete test.

BUILD NOTES

Test entities are visibly marked and never saved. Debug data is permission-gated and streamed only while this panel is open.

DEBUG & VALIDATION / HOLLOW KING

2 PROBLEMS

CLOSE

RULE TRACE / PHASE TWO

TRIGGER

Health Changed

CONDITION

Health <= 50%

CONDITION

phase == 1

ACTION

Set phase = 2

RULE DID NOT EXECUTE

phase was 2; expected 1

VALIDATION

ERROR

GO TO >

Missing texture

bandti.png

WARNING

GO TO >

Melee combat has no target selector

AI / Targeting

WARNING

GO TO >

Boss bar with only 12 max health

Attributes

CLOSE

SAVE

INTENT

Explain why logic did not run and centralize structural or logical problems before save.

KEY STATES

Passed/failed condition, trigger not fired, unknown node, warning vs error, hidden advanced error, all clear.

BUILD NOTES

Debug traces are transient. Validation issues keep stable focus targets so GO TO opens the responsible control or node.

REFERENCES TO mypack:bandit

5 TOTAL

CLOSE

QUESTS

2

Clear Bandit Camp

clear_bandit_camp

OPEN >

Road to Village

road_to_village

OPEN >

SPAWN TABLES

1

Forest Bandits

forest_bandits

OPEN >

DIALOGUE

1

Prisoner Dialogue

prisoner_dialogue

OPEN >

ENTITY

1

Bandit Chief

bandit_chief

OPEN >



DELETE BANDIT?

This definition has 5 references

and 12 placed instances.

DELETING MAY BREAK

quests and spawn tables

placed instances become orphaned

JSON is backed up

MIGRATE REFERENCES

CANCEL

DELETE ANYWAY

INTENT

Show where definitions are used and make destructive changes proportional to their impact.

KEY STATES

Grouped references, stale reference, zero references, open target, delete anyway, migration, unique instances.

BUILD NOTES

Deleting a definition never silently deletes placed instances. Offer migration or explicit orphan handling where supported.

EDITOR COMPLEXITY

BASIC

● GENERAL

● APPEARANCE

● ATTRIBUTES

● MOVEMENT

● AI

● COMBAT

● EQUIPMENT

● INTERACTION

● DROPS

ADVANCED

● VARIABLES

● RULES

● SCHEDULES

● SPAWN CONDITIONS

● CUSTOM ATTACKS

● INSTANCE OVERRIDES

● RAW JSON

● RUNTIME DEBUG

ESSENTIAL SHORTCUTS

CTRL+S

Save

CTRL+Z

Undo

CTRL+Y

Redo

TAB

Add node

F

Focus

ESC

Cancel tool

INTENT

Let beginners build a complete enemy or NPC without hiding existing advanced data or save errors.

KEY STATES

Mode switch, hidden-page blocker, first-run help, per-user preference, addon component, shortcut overlay.

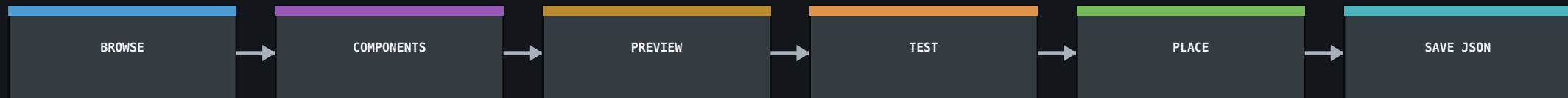
BUILD NOTES

Mode changes visibility only. Saving in Basic mode preserves variables, rules, schedules, attacks, and unknown fields.

COMPLETE ENTITY WORKFLOW

ONE BASE ENTITY. AN ENTIRE WORLD OF CHARACTERS.

The editor keeps a blacksmith simple, gives a bandit complete combat behavior, and lets a boss grow into variables, phases, custom attacks, and debug traces - all through the same JSON definition system.



SOURCE OF TRUTH

`data/mypack/rpg/entities/village_guard.json`